

**Tollbooth Civilization:
Why the Work That Becomes Foundational
Cannot Be Owned, and What Gets Owned Instead**

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Abstract

Rankings of private wealth are commonly read as rankings of contribution; the error has a definite structure. Work becomes foundational by becoming non-excludable, standardized, cheap to copy, and depended upon by everything built above it, and these same properties prevent its creator from charging for it. The designers of the internet's transport protocol, the metal-oxide-semiconductor transistor, the SQLite database, and oral rehydration therapy captured almost none of the value their work supports. We model a technological civilization as a stack of serially dependent layers, each able to levy a per-use toll. In a normalized market the open stack delivers social value 0.5 and its creators capture nothing; a single owner of the whole stack captures 0.25 and halves throughput; two independent tollbooths set a higher combined price than a single owner, collect less between them, and leave users worse off; twenty tollbooths reduce throughput to about 5 percent of the open level. When all layers charge a common revenue-maximizing toll, the capturable share of any one layer falls as $1/(2N)$ in the number of complementary layers, reaching the 2.2 percent of social surplus that Nordhaus estimated innovators capture at a depth of about 23 layers. Social value is maximized when creators capture nothing. Large fortunes form instead at narrower, excludable layers above the commons, where control of a point that use must pass through yields recurring revenue, and that control is then described as authorship of the infrastructure beneath it. Appropriability still funds invention, integration is real work, and a single integrated owner does less damage than a fragmented anticommons.

Introduction

In 1974 Vint Cerf and Robert Kahn published a design for moving data between networks that had no knowledge of one another, and they charged nothing for it (Cerf and Kahn 1974). Suppose they had attached a toll of a hundredth of a cent, 10^{-4} dollars, to each packet crossing the boundary their protocol defined. Their revenue would be that toll multiplied by the number of packets. The number cannot be audited, but the internet has carried on the order of 10^{18} to 10^{19} packets, and a hundredth of a cent on 10^{19} packets is 10^{15} dollars, a quadrillion.

The figure is absurd, and the absurdity illustrates the problem. A per-packet toll high enough to make Cerf rich would have removed the incentive to build the internet. Every device that speaks the protocol does so because adoption costs nothing and requires no one's permission. A turnstile at the boundary would cause traffic to route around it or never to form. The value of TCP and the impossibility of charging for TCP are the same fact. The protocol acquired its enormous value by becoming something no one had to pay to use.

The same pattern recurs across foundational technologies. The metal-oxide-semiconductor field-effect transistor, demonstrated at Bell Labs in 1959 by Mohamed Atalla and Dawon Kahng, is the switching element of essentially all digital electronics; the Computer History Museum describes the MOS transistor as the most frequently manufactured object in human history, with cumulative production around 10^{22} devices. SQLite, written and placed in the public domain by D. Richard Hipp, runs inside most phones and browsers and, by its authors' count, over a trillion

active databases. Oral rehydration therapy is water, salt, and sugar in proportions that exploit the coupled transport of sodium and glucose across the gut wall; in 1978 *The Lancet* judged the underlying discovery potentially the most important medical advance of the century, and no fortune has been built on the correct ratio. By the ordinary measure, the people who made these things are not rich, and the ordinary measure is what requires examination.

A wealth ranking orders people by the market value of the claims they own. That value tracks the revenue a claim can yield, which tracks how effectively others can be excluded from what the claim controls, which bears little relation to how much the underlying thing contributed to anyone's life. Nordhaus (2004) measured the gap directly. Estimating the share of the social surplus from technological advance retained by its producers in the United States nonfarm business sector from 1948 to 2001, he found that innovators capture about 2.2 percent, with the remaining surplus flowing to consumers and to later producers who build on the advance. His abstract calls the captured share a minuscule fraction. Against a list of the wealthiest people, the finding implies more than modest underpayment: the quantity a wealth ranking measures and the quantity it is taken to measure differ, and the second exceeds the first by a factor of more than 40.

Economics of non-appropriability

For the most foundational work, the low capture rate is a structural consequence of what foundational means, and no market adjustment will remove it. The properties that make a piece of work foundational are the properties that make it uncapturable.

Information is the base case, stated by Arrow (1962) as a paradox. Technical knowledge is non-rival, so one person's use does not diminish another's, and it is non-excludable once disclosed: to sell it, the seller must reveal it, and once revealed it can be used without further payment. Samuelson (1954) had stated the general form eight years earlier. A pure public good is one whose consumption by one agent subtracts nothing from its availability to others, and unregulated markets underprovide it because no seller can meter what no one can be excluded from. Romer (1990) based growth theory on the same property, treating ideas as non-rival inputs whose free reuse drives long-run growth because their value is not exhausted by their first holder. The more completely a thing has the character of knowledge, the less of the value it generates can be enclosed.

Foundational technology inherits this property and adds three more. It is standardized, so its value derives from everyone using the same version, and a toll that fragments adoption destroys the standardization that produced the value. It is cheap to replicate, so any price much above zero invites a free substitute, which need only be unowned to compete. And it is complementary to final products, delivering its value inside products that thousands of other firms assemble and sell, so the surplus appears on their books and in their customers' lives and never passes a point of sale the originator controls. Teece (1986) characterized the resulting regime. When appropriability is weak, the returns from an innovation flow to the owners of the complementary assets it needs to reach a market, namely manufacturing, distribution, brand, and installed base, and the inventor of a widely usable primitive owns none of these by virtue of the invention.

The base of the stack also tends to have no owner because many people built it. Allen (1983) documented collective invention among the ironmasters of the Cleveland district, who published their blast-furnace dimensions and yields to one another and produced a rapid cumulative advance that belonged to no single firm. Much foundational work has this form. TCP became TCP/IP and then a protocol suite maintained by a standards community; the transistor passed through Bell Labs and a thousand later refinements; a protocol gains authority by being implemented independently many times. When a capability is produced collectively, the question of who should capture its value has no clean answer.

The argument turns on the ratio of two quantities. Let S_i be the social value a contribution i creates, the amount by which the world is better with it than with its most likely absence, and P_i the private value its creator can capture. Define the capture ratio and its inverse, civilizational leverage,

$$\kappa_i = \frac{P_i}{S_i}, \quad L_i = \frac{S_i}{P_i} = \frac{1}{\kappa_i}.$$

Contribution is the numerator of L_i and appropriation its denominator. For the most foundational work the two are inversely related: non-excludability, standardization, replicability, and complementarity raise S_i and push P_i toward zero. A wealth ranking orders the world by P_i and is read as if it ordered by S_i , so the people with the highest L_i do not appear in it.

The layered stack and its chokepoints

A technological civilization is a stack of layers that depend on one another in series. Mathematics, measurement, and published science form the base. Above them are technical primitives: the transistor, the transport protocol, the name system, the clock-synchronization protocol, and compression and cryptographic routines. Above those are shared infrastructures such as grids, networks, fabrication plants, and public research institutions. Above those are platforms and production systems, including operating systems, clouds, design suites, and app stores. Near the top are consumer products and brands, and at the top the financial and narrative layer: equity, and the account of who made the whole. Each layer runs on those below and is unaware of most of them.

Dependency and revenue flow through the stack in opposite directions. Dependency flows down: every layer needs those beneath it, and the lower a layer sits, the more of the structure depends on it. Money and recognition flow up: the higher a layer sits, the closer it is to a paying customer, a billing relationship, a proprietary account, a capital market, and a name a journalist can attach to the whole. The layer on which most depends and the layer that captures most revenue lie near opposite ends of the same column. Bresnahan and Trajtenberg (1995) named the lower layers general-purpose technologies, defined by pervasiveness across sectors, continual improvement, and innovational complementarities that seed advances in every field they touch. Generality and appropriability are opposed: a technology qualifies as general-purpose by being useful everywhere, which makes it ownable nowhere in particular.

A tollbooth forms where a layer can be made both narrow and unavoidable. Narrowness comes from a proprietary format, an interface, a certification, a distribution channel, a payment rail, a store, or a standard-essential patent. Unavoidability comes from lock-in. David (1985) derived the mechanism from the QWERTY keyboard, which persists because everyone else uses it and the cost of coordinating a switch exceeds any one typist's gain, and whose speed is irrelevant to its persistence. Where a layer combines narrowness and unavoidability, its owner is paid for occupying a point that traffic cannot bypass, with no ongoing production required. Christophers (2020) describes the contemporary economy as increasingly organized around ownership of scarce assets such as land, spectrum, intellectual property, platforms, and contracts, which yield income from control. Giblin and Doctorow (2022) trace the same structure through the creative industries, where a few firms occupy the chokepoint between creators and audiences and extract from both. A tollbooth is a priced chokepoint.

A model of serial tolls

The proposal that a foundational creator should charge a small fee fails on arithmetic. If a use of the stack requires many layers in series and each layer's owner sets a toll to maximize that layer's revenue, the tolls interact to the detriment of everyone, owners included. The result dates to Cournot. In a chapter on producers of complementary goods first published in 1838, he considered copper and zinc, each held by a separate monopolist and combined into brass, and showed that two independent owners set a higher total price and sell less than a single owner of both metals would (Cournot [1838] 1897). The accompanying model applies this result to the technological stack.

Normalize the market to a unit mass of potential uses (packets, transactions, or new products), each with willingness to pay drawn uniformly on the interval from zero to one, and no cost of provision beyond the tolls, so that demand for the stack at total per-use price P is $Q(P) = 1 - P$. Let the stack have N essential layers in series, each owned by a party that sets a per-use toll r_i to maximize $r_i Q$, taking the others as given. The symmetric Nash equilibrium and its welfare are

$$r_i^* = \frac{1}{N+1}, \quad P_N = \frac{N}{N+1}, \quad Q_N = \frac{1}{N+1}, \quad W_N = \frac{2N+1}{2(N+1)^2},$$

where W_N is the sum of consumer surplus and total toll revenue. Three ownership structures of the same stack can then be compared.

The open stack charges no tolls. Its throughput is the entire market and its welfare is 0.5, all consumer surplus and none captured. A single owner of the whole stack sets one price, captures 0.25, and reduces throughput to 0.5 and welfare to 0.375; capturing a quarter destroys an eighth of the surplus, the standard deadweight loss of monopoly. Dividing that owner into independent tollbooths introduces the Cournot interaction. Two tollbooths on the same road each charge a third, for a combined price of 0.667, above the 0.5 a single owner would set. Throughput falls to 0.333. The two tollbooths together collect 0.222, less than the 0.25 of a single owner, and consumer sur-

plus falls from 0.125 to 0.056. Dividing one road between two tollbooths makes both users and owners poorer. This is the tragedy of the anticommons identified by Heller and Eisenberg (1998) in biomedical research, where too many holders of exclusion rights leave a resource underused because each can block the others; here it appears in price form.

The collapse accelerates with depth. Ten independent tollbooths reduce throughput to about 9 percent of the open level and twenty to about 5 percent. Welfare falls to about a sixth of its open value at ten tollbooths and below a tenth at twenty. Twenty foundational layers, each levying an individually trivial toll, do not enrich twenty inventors; they suppress the throughput that generated the value, and the suppression worsens with every added layer. Shapiro (2001) described the applied version as the patent thicket, the dense overlap of complementary rights a product must clear before it can ship, and Lemley and Shapiro (2007) named the pricing pathology royalty stacking, the accumulation of individually reasonable royalties into an unpayable total. The magnitude in real standard-essential-patent disputes is contested, but the model's direction holds: independent tolls on complements accumulate faster than the market can bear.

The same model gives a limit on appropriation. Consider one foundational layer in a stack of N complements and ask how much its owner can charge without collapsing adoption. If all N layers charge a common toll r , demand is $Q = 1 - Nr$ and each owner's revenue $r(1 - Nr)$ is maximized at $r^* = 1/(2N)$, which divides the integrated-monopoly price equally among the layers and yields a per-layer revenue of $1/(4N)$. Crediting the layer with the entire open surplus of 0.5, for which each perfect complement is a necessary condition, the capture ratio and leverage are

$$\kappa(N) = \frac{1}{2N} \rightarrow 0, \quad L(N) = 2N \rightarrow \infty \quad (N \rightarrow \infty).$$

The deeper the stack above a layer, the smaller the share of the value passing through it that its owner can take. In a stack 20 layers deep, the sustainable capture of any single foundational layer is 2.5 percent, a civilizational leverage of 40 to 1. The model's capture ratio equals Nordhaus's empirical 2.2 percent at a depth of about 23 layers (22.7 in the continuous formula). This is not a calibration, although the coincidence between a stipulated model and a measured economy is notable. Greater foundationality lowers what a creator can charge, toward zero.

Two checks test the model. First, the orderings do not depend on linear demand. Under the family $Q(P) = (1 - P)^\theta$, the symmetric equilibrium toll has the closed form $r^* = 1/(N + \theta)$, and both the throughput collapse and the welfare ranking of open above integrated above fragmented hold for every positive θ . Across 2,000 seeded draws of curvature and market size, every draw preserves both orderings, while the depth of the collapse varies: throughput with ten tolled layers is between about 4 and 18 percent of the open level, depending on curvature. Second, the single integrated owner is a genuine second-best. A monopolist owning the entire stack internalizes the complementarity and prices it once, doing strictly less damage than the fragmented anticommons. Openness ranks first, integration second, and fragmentation last, so the model does not establish that all ownership is illegitimate.

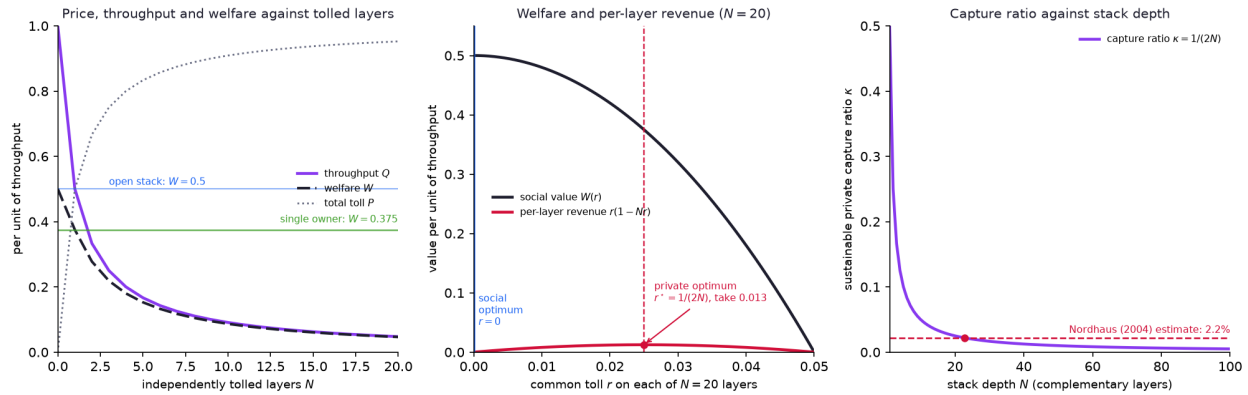


Figure 1: Tolls on a serially dependent stack in the normalized model. Left: total toll P , throughput Q , and welfare W as the number of independently tolled layers N increases; two tollbooths already price above the single-owner benchmark (welfare 0.375), and both lie below the open stack (welfare 0.5). Centre: for a stack of $N = 20$ layers charging a common toll r , social value $W(r)$ is maximal at $r = 0$, while per-layer revenue peaks at $r^* = 1/(2N)$ with a take of 0.0125. Right: the capture ratio $\kappa = 1/(2N)$ against stack depth, crossing the 2.2 percent of social surplus estimated by Nordhaus (2004) at a depth of about 23. All values are properties of the model.

Capture at the upper layers and its attribution

The fortunes the stack generates are real, and they accumulate at the excludable layers near its top. The significant step is in the attribution: a claim on one layer is routinely presented as authorship of the entire column beneath it.

The mechanism is visible where the public record is unusually complete. NASA's Commercial Orbital Transportation Services program was designed to create a private launch industry, paying milestone-based awards, including a few hundred million dollars to SpaceX, with the agency and its space station as anchor customer and decades of accumulated public science as the substrate. SpaceX then performed real and difficult engineering, reduced launch costs, and outperformed incumbent contractors. Both statements hold. The popular account retains only the second, reducing a distributed system of public research, public procurement, institutional knowledge, suppliers, workers, and private engineering to the name and personality of the residual equity holder. Mazzucato (2013) assembled evidence that the state routinely occupies the foundational, risk-absorbing layers of this kind, from the internet's protocols to the components of a smartphone to the frontier of pharmaceutical research, and Mazzucato (2018) described the general operation as the presentation of value extraction as value creation. Ownership of a layer is recast as construction of the stack.

In software the reattribution is least conspicuous. A design-software company does not own geometry, the calculus, or the numerical methods its product runs on; these belong to the epistemic commons and cannot be enclosed. It can own the proprietary format in which a firm's accumulated work is stored, the interface others must build against, and the account through which collaboration flows. It converts universally available mathematics into a non-portable operational depen-

dency, so that a firm's continued access to its own past work depends on its continued relationship with a vendor. The value captured is the switching cost built around the mathematics. The same industry shows that the reverse is possible: in 2008 Adobe placed the core of the PDF specification into an open international standard, ISO 32000-1, and issued a royalty-free patent licence for it, making the format more infrastructural by making it less owned and leaving its proprietary application business to compete above the open layer. A layer and the enclosure of that layer are separable, and the relevant question is which of the two a firm is paid for.

Rent-seeking adds a cost that a wealth ranking cannot register. Tullock (1967) observed that the social cost of monopoly includes, beyond the deadweight triangle, the real resources spent acquiring and defending the monopoly position: legal and lobbying expenditure, and effort that competes for a rent without producing anything. Krueger (1974) generalized this into the political economy of the rent-seeking society, in which control over access, whether an import licence, a spectrum grant, or a standard, becomes a prize that justifies expending real resources. The tollbooth economy depends on this expenditure. Much of what the upper layers spend goes to securing and publicizing positions on existing infrastructure, and that spending, where counted at all, is recorded as the cost of creating value it only captures.

Case studies

The pattern holds in particular cases when the counterfactual is stated carefully. The value of a foundational contribution excludes the value of everything built on it, which belongs partly to the builders, and includes the acceleration, cost reduction, standardization, and interoperability it provided over the next-best alternative. Even under this narrow measure, the leverage is extreme and capture is near zero.

The transport protocol is the clearest case. Cerf and Kahn published TCP openly, took no royalty, and shared the 2004 Turing Award for it. The counterfactual is a world in which internet-working arrived later, more fragmented, and less interoperable, and the value of averting that reached nearly everyone except the authors. The transistor is the largest in magnitude. Atalla and Kahng's surface-passivation work made the MOS transistor manufacturable, and after roughly 10^{22} units their names remain unknown to a public that carries billions of their descendants in its pockets. SQLite is the purest case of deliberate non-appropriation: a concentrated authorship, near-universal deployment, and a public-domain dedication that forecloses capture by design, so that a database present in more devices than any other returns to its author a salary and a reputation, neither of which scales with use. Oral rehydration therapy is the case most damaging to the wealth measure: a discovery credited with saving tens of millions of lives, unpatentable because it is a ratio of three cheap substances. The formula costs pennies, and on any balance sheet that records ownership its value is zero.

Open-source software shows the divergence in a single pair of numbers. Hoffmann et al. (2024) estimated the cost of rebuilding the freely available software on which firms depend and the amount firms would have to pay to replace it, finding a supply-side cost near 4.15 billion dollars and a demand-side value near 8.8 trillion, a ratio above a thousand between the value the software deliv-

ers and the cost of the labour that produced it. The demand-side figure is a modelled counterfactual and should be read as an estimate, but its structure supports the argument: the same code, copied at no cost into thousands of firms, generates value far exceeding its production cost and accruing to none of its producers. Informal figures point the same way and require more caution. A figure of 15.85 trillion dollars circulates for the lifetime consumer savings from one power-switching device, the insulated-gate bipolar transistor; it comes from a compendium assembled by the inventor, rests on assumptions such as 50 percent global penetration of variable-speed drives, and is not an independent measurement. Its appropriate use is to note that its order of magnitude is unremarkable in this domain and that the person associated with it captured a negligible fraction of it.

Scope and limits of the argument

The claim is a systematic divergence between contribution and capture, together with a habit of presenting control as creation. It does not assert that all wealth is rent, that appropriation is theft, or that those at the top of the stack contribute nothing. Each of those claims would be false, and the model itself shows why.

Appropriability has real effects. The prospect of capturing some of an innovation's value is an incentive to produce it, and where imitation is easy and development expensive, a period of exclusion may be the only source of development funding. The 2.2 percent Nordhaus measured is small but positive, and at the margin it affects invention. The public-goods tradition from Samuelson through Arrow to Romer treats under-appropriation as an under-provision problem for this reason. Driving every capture ratio to zero would reduce the supply of foundational work, so two propositions must be held together: appropriation motivates creation, and appropriation and creation are distinct quantities that diverge most for the most important work.

Integration and scale are also real. Turning a set of primitives into a product a person can buy and rely on is difficult and valuable work, and firms that do it are not illegitimate for depending on public and open foundations, since every firm does. Teece's analysis (1986) applies in both directions. Capture of an innovation's value by the owner of complementary assets is sometimes rent extracted from an inventor and sometimes a fair return to the hard work of manufacturing, distributing, and supporting a product at scale. The model also shows that a single integrated owner is preferable to the anticommons, so dividing the upper layers into competing tollbooths would worsen the problem.

The analysis supports narrower conclusions. A wealth ranking should be read as a ranking of appropriability and bottleneck control, distinct from a ranking of contribution. Accounts that attribute a distributed, publicly seeded technological system to a single founder warrant scepticism. Policy should favour keeping foundational layers open, collect public returns at the level of the firm or portfolio in preference to a tax on each use of a primitive, and distinguish the second-best of integration from the creation of the underlying layers. Above all, a position on existing infrastructure should not be accepted as evidence of having built it.

Conclusion

The economics of information has long described non-appropriability as a market failure, a case in which private incentives fall short of the social optimum. On foundational layers, non-appropriability is instead a condition for the layer to function at all.

In the model, social value of the stack is maximized in the open configuration, where throughput is complete, welfare is 0.5, and creators capture nothing; every move toward capture, by one integrated owner or by many independent tollbooths, strictly lowers welfare. Producers retain little of the surplus, and the remainder, about 98 percent, reaches everyone else, which is the form in which a foundational technology delivers value. A metered transport protocol would not have become the internet; a transistor carrying a per-unit royalty to its inventor would not have been manufactured 10^{22} times; a licensed rehydration formula would not have reached the children it reached.

Cerf's unissued invoice is therefore the mechanism by which the protocol's value was delivered, and the reason a network exists on which tolls can be attempted. A society that ranks by capture assigns its largest rewards and highest status to the layers that charge and little recognition to the layers that carry the load. The appropriate response is to distinguish control of a chokepoint from construction of the infrastructure behind it, and to recognize that much of that infrastructure was built by people who could not, and did not, charge for it.

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